

Relational Database Design & Schema Normalization

Entity-Relationship Modeling, Constraints, and Online Bookstore Schema Architecture

TRAINEE	MENTOR	INSTITUTION	DATE & STATUS
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1. EXECUTIVE SUMMARY

This technical report documents the advanced database modeling, normalization strategies, and schema creation completed during Day 12 of the Database Development phase. Moving beyond the flat-file table configurations established previously, the focus of this session was to engineer a fully relational, multi-entity blueprint using Entity-Relationship (ER) modeling techniques. To prevent update anomalies, structural data redundancy, and broken relational references, a decoupled relational schema was engineered for an enterprise-level Online Book Store. The production model was systematically refactored to comply with Third Normal Form (3NF) requirements.

2. ADVANCED NORMALIZATION MECHANICS & COMPLEXITY ARCHITECTURE

To prevent structural data loss common in volatile program states, core memory paradigms were evolved into disk-based relational storage vectors:

- **First Normal Form (1NF):** Eliminated repeating column groups and multi-valued composite attributes, ensuring that every intersecting cell contains strictly atomic values.
- **Second Normal Form (2NF):** Verified full functional dependencies by confirming all non-prime attributes depend entirely on the composite primary keys, stripping out partial dependencies.
- **Third Normal Form (3NF):** Removed hidden transitive functional dependencies by splitting out non-key parameters that relied on intermediate fields rather than directly on the primary key, achieving stable constant lookups, **O(1)**.

3. STRUCTURAL MAPPING MATRIX

RELATIONAL ENTITY	KEY CONSTRAINTS MAPPING	CARDINALITY ASSOCIATION	NORMALIZATION STATUS
Customers	customer_id (PK)	1 : N with Orders	3NF Compliant
Books	book_id (PK), author_id (FK)	N : 1 with Authors	3NF Compliant
Orders	order_id (PK), customer_id (FK)	1 : N with Order_Details	3NF Compliant
Order_Details	(order_id, book_id) (Composite PK)	Associative Entity Mapping (M:N)	3NF Compliant

4. TECHNICAL DATABASE IMPLEMENTATION

The following production-ready DDL script establishes the relational schema constraints, primary identifiers, and cascading index relationships required for the bookstore application:

```

-- 1. Initialize and Secure Isolated Database Domain
CREATE DATABASE IF NOT EXISTS OnlineBookStore;
USE OnlineBookStore;

-- 2. Build Independent Parent Entities (3NF Master Catalogs)
CREATE TABLE authors (
    author_id INT AUTO INCREMENT,
    author_name VARCHAR(100) NOT NULL,
    PRIMARY KEY (author_id)
);

CREATE TABLE customers (
    customer_id INT AUTO_INCREMENT,
    first_name VARCHAR(50) NOT NULL,
    email VARCHAR(100) UNIQUE NOT NULL,
    PRIMARY KEY (customer_id)
);

-- 3. Build Dependent Child Entities (With Referential Constraints)
CREATE TABLE books (
    book_id INT AUTO_INCREMENT,
    title VARCHAR(150) NOT NULL,
    author_id INT,
    price DECIMAL(8, 2) CHECK (price >= 0),
    stock_quantity INT DEFAULT 0,
    PRIMARY KEY (book_id),
    FOREIGN KEY (author_id) REFERENCES authors(author_id)
);

CREATE TABLE orders (
    order_id INT AUTO_INCREMENT,
    customer_id INT,
    order_date TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    PRIMARY KEY (order_id),
    FOREIGN KEY (customer_id) REFERENCES customers(customer_id)
);

-- 4. Build Associative Table to Resolve Many-to-Many Relationships
CREATE TABLE order_details (
    order_id INT,
    book_id INT,
    quantity INT CHECK (quantity > 0),
    price_at_purchase DECIMAL(8, 2),
    PRIMARY KEY (order_id, book_id),
    FOREIGN KEY (order_id) REFERENCES orders(order_id) ON DELETE CASCADE,
    FOREIGN KEY (book_id) REFERENCES books(book_id)
);

```

5. WORKSPACE ENVIRONMENT STATE MOCKUPS

RELATIONAL ENTITY-RELATIONSHIP CONNECTIVITY MAP (LOGICAL LAYOUT)

[authors] 1 ----- M [books] 1 ----- M [order_details] M ----- 1 [orders] M ----- 1 [customers]

* Relational Interlocks:

- authors.author_id mappings securely link book metadata categories.
- order_details serves as the cross-cutting associative bridge resolving (M:N) book purchases.

VERSION CONTROL TRACK METRICS (GITHUB SUBMISSION REPOSITORY STATE)

```
Origin: remote upstream master branch
[Commit Ref: #DB-D12-N41] feat: design conceptual ER diagram layout mappings for bookstore
schema
[Commit Ref: #DB-D12-N42] feat: refactor storage schemas down to 3NF parameters to eliminate
update anomalies
[Commit Ref: #DB-D12-N43] feat: implement associative order_details mapping table with
cascading purges
```

6. LEARNING OUTCOMES

- **ER Diagram Formulation:** Mapped conceptual bookstore domain criteria into distinct relational entities, attributes, and cardinality boundaries.
- **Data Structural Normalization:** Refactored repeating groups, partial factors, and transitive links to satisfy full 3NF operational safety standards.
- **Associative Mapping Resolution:** Deployed customized linking entities with compound keys to smoothly resolve complex Many-to-Many entity connections.
- **Cascading Rule Configuration:** Set up robust foreign keys and cascading delete constraints to prevent isolated orphan rows from compromising data consistency.